

MATH 55 FIRST MIDTERM EXAM, PROF. SRIVASTAVA
SEPTEMBER 29, 2022, 5:10PM–6:30PM, 2050 VLSB.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	12
2	8
3	5
4	8
5	8
6	9
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation.

(a) (3 points) The compound propositions **T F**

$$\forall x \exists y (P(x) \wedge Q(y))$$

and

$$\exists x \forall y (\neg P(x) \vee \neg Q(y))$$

are negations of each other.

(b) (3 points) The compound proposition **T F**

$$(\exists y \forall x (P(x, y))) \rightarrow (\forall x \exists y (P(x, y)))$$

is a tautology.

(c) (3 points) If $f : A \rightarrow B$ and S and T are subsets of A then **T F**

$$f(S - T) \supseteq f(S) - f(T).$$

(where $f(S) = \{b \in B : \exists s \in S \text{ s.t. } f(s) = b\}$ denotes the image of a subset of A).

(d) (3 points) If $d, x, y, \in \mathbb{Z}_+$ and $d|xy$ then $d|x$ or $d|y$. **T F**

[Scratch Space Below]

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2. (a) (4 points) **Prove or Disprove** the following statement: if $f : B \rightarrow C$ and $g : A \rightarrow B$ are functions and $f \circ g$ is one-to-one, then g must be one-to-one.

- (b) (4 points) **Prove or Disprove** the following statement: if $f : B \rightarrow C$ and $g : A \rightarrow B$ are functions and $f \circ g$ is one-to-one, then f must be one-to-one.

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3. (5 points) Prove that if A is a countably infinite set and B is a finite set, then $A \times B$ is countably infinite.

[Scratch Space Below]

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4. (a) (4 points) Prove that if x and y are rational numbers such that $x < y$ then there exists a rational number z such that $x < z < y$.

- (b) (4 points) Prove that if x and y are rational numbers such that $x < y$, then the set

$$S = \{z \in \mathbb{Q} : x < z < y\}$$

is infinite (here $\mathbb{Q}$ denotes the set of rational numbers. hint: use your solution to the previous part of the question.).

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5. (a) (4 points) Calculate

$$(38)^{2022} \bmod 5.$$

- (b) (4 points) Prove that there do not exist integers $a, b, c \in \mathbb{Z}$ such that

$$a^8 + b^8 + c^8 = 555554.$$

(hint: consider the remainder modulo 5 of both sides of this equation)

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[Scratch Paper 1]

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6. (a) (3 points) Use the Euclidean algorithm to calculate $\gcd(9, 44)$.

(b) (3 points) Use your answer to part (i) to find an integer $x \in \{0, 1, \dots, 44\}$ which satisfies

$$9x \equiv 5 \pmod{44}.$$

(c) (3 points) Prove that there is no integer $x \in \mathbb{Z}$ which satisfies the congruence

$$8x \equiv 5 \pmod{44}.$$

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[Scratch Paper 2]

Name and SID: _____

[Scratch Paper 3]